PSGR Krishnammal College for Women

CLOUD-BASED SIEM ANALYTICS AND AUTOMATED INCIDENT RESPONSE USING PYTHON

INTERNSHIP WORK SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF COMPUTER SCIENCE WITH CYBER SECURITY

SUBMITTED

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JULY 2026

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ABSTRACT

The increasing number of cyber threats has made continuous security monitoring an essential requirement for organizations. Security Operations Centers (SOCs) rely on Security Information and Event Management (SIEM) systems to collect, analyze, and correlate security events from multiple sources. However, traditional monitoring methods often require significant manual effort, making it difficult to detect and respond to threats efficiently. This project proposes a Cloud-Based SIEM Analytics and Automated Incident Response system that integrates Python automation with the Sumo Logic cloud platform to enhance security monitoring and incident management.

The proposed system simulates various cyber attack scenarios by generating security logs through Python. These logs are enriched using Threat Intelligence, GeoIP information, MITRE ATT&CK technique mapping, and risk scoring to provide meaningful security insights. The processed logs are forwarded to Sumo Logic, where interactive dashboards visualize security events in real time. In addition, the system automatically generates executive reports, incident reports, IOC summaries, SOC email alerts, and firewall blocking actions to demonstrate automated incident response capabilities.

The implementation improves centralized log monitoring, threat detection, attack classification, and security event visualization within a cloud-based SOC environment. By reducing manual analysis and providing real-time security insights, the proposed system demonstrates an effective approach to modern security operations. The project highlights how cloud-based SIEM solutions can improve threat visibility, support faster incident response, and strengthen organizational cybersecurity.

I.INTRODUCTION

In today's digital era, organizations increasingly depend on information technology and cloud computing to manage their operations and protect critical business assets. As digital transformation continues to expand, the volume of security logs generated from servers, applications, firewalls, and network devices has grown significantly. At the same time, cyber threats such as malware, phishing, ransomware, brute-force attacks, SQL injection, and network intrusions have become more sophisticated, making effective security monitoring a major challenge for organizations.

Traditional security monitoring methods often rely on multiple security tools and manual analysis of security logs. Due to the large volume of security events generated every day, identifying malicious activities manually becomes difficult, leading to delayed threat detection and slower incident response. To overcome these limitations, organizations are increasingly adopting Security Information and Event Management (SIEM) systems that centralize log collection, correlate security events, detect suspicious activities, and provide real-time visibility into the organization's security posture. Cloud-based SIEM platforms further enhance these capabilities through scalable infrastructure, automated analytics, interactive dashboards, and centralized monitoring.

This project, "Cloud-Based SIEM Analytics and Automated Incident Response Using Python," presents a cloud-based security monitoring solution by integrating Python automation with the Sumo Logic SIEM platform. The proposed system simulates cyber attack scenarios, enriches security logs using Threat Intelligence, GeoIP information, MITRE ATT&CK mapping, and risk scoring, and visualizes security events through interactive dashboards. In addition, it automates incident response by generating Executive SOC Reports, Incident Reports, IOC summaries, SOC email alerts, and firewall blocking actions. The project demonstrates how cloud-based SIEM combined with Python automation can improve threat detection, centralized monitoring, and incident response while providing practical exposure to modern Security Operations Center (SOC) workflows.

1.1 ORGANIZATION PROFILE

PSGR Krishnammal College for Women, Coimbatore, is one of the premier autonomous institutions in Tamil Nadu, committed to providing quality higher education and fostering academic excellence. The college offers a wide range of undergraduate, postgraduate, and research programs across multiple disciplines, emphasizing innovation, skill development, and industry-oriented learning. With well-equipped laboratories, experienced faculty members, and a student-centric learning environment, the institution continuously promotes research, technological advancement, and professional growth among students.

The Department of Computer Science with Cyber Security aims to equip students with strong theoretical knowledge and practical expertise in emerging areas of cybersecurity. The department offers specialized training in ethical hacking, network security, digital forensics, cloud security, web application security, and Security Operations Center (SOC) practices. Through workshops, certification programs, internships, and real-time projects, students gain practical exposure to modern cybersecurity tools and technologies, enabling them to address real-world security challenges effectively.

This project, "Cloud-Based SIEM Analytics and Automated Incident Response Using Python," was carried out as part of the academic curriculum under the Department of Computer Science with Cyber Security at PSGR Krishnammal College for Women. The project focuses on implementing a cloud-based Security Information and Event Management (SIEM) solution using Python and Sumo Logic to automate log analysis, threat intelligence integration, incident response, and security monitoring. It reflects the department's commitment to bridging the gap between theoretical concepts and practical implementation by encouraging students to develop innovative solutions for current cybersecurity challenges.

1.2 PROBLEM IDENTIFICATION

The rapid growth of digital technologies has significantly increased the volume of security events generated by servers, applications, network devices, and cloud environments. Organizations continuously face cyber threats such as brute-force attacks, SQL injection, phishing, malware, ransomware, unauthorized access, and reconnaissance activities. Monitoring these security events manually has become increasingly difficult due to the massive amount of log data generated every day. As a result, security analysts may overlook critical threats, leading to delayed incident detection and slower response times.

Many organizations still depend on traditional security monitoring methods that use isolated security tools and manual log analysis. These approaches lack centralized visibility, automated event correlation, and real-time threat detection. The absence of threat intelligence integration, risk prioritization, and standardized attack mapping makes it difficult for Security Operations Centers (SOCs) to identify genuine threats among thousands of daily security events. This increases the workload of security teams and reduces the overall efficiency of incident response.

The primary challenge is to develop a centralized cloud-based security monitoring system that can automatically collect, analyze, enrich, and visualize security events while supporting rapid incident response. Therefore, this project proposes a Cloud-Based SIEM Analytics and Automated Incident Response system that integrates Python automation with the Sumo Logic cloud SIEM platform. The proposed system enhances security monitoring by incorporating threat intelligence, MITRE ATT&CK mapping, GeoIP analysis, risk scoring, automated reporting, and real-time dashboards to improve threat detection and support efficient SOC operations.

II.LITERATURE SURVEY

S.No	TITLE	AUTHOR	METHODOLOGY	TOOLS USED	KEY FINDINGS
1	Security Information and Event Management (SIEM): A Survey	M. Almseidin et al.	Surveyed SIEM architectures and security monitoring techniques.	SIEM Platforms	SIEM improves centralized log management, threat detection, and incident response.
2	MITRE ATT&CK: Design and Philosophy	MITRE Corporation	Introduced the ATT&CK framework for mapping adversary tactics and techniques.	MITRE ATT&CK Framework	Standardizes attack classification and enhances threat investigation.
3	Threat Intelligence in Cyber Security	SANS Institute	Discussed the use of threat intelligence for identifying malicious activities and Indicators of Compromise (IOCs).	Threat Intelligence Feeds	Threat intelligence significantly improves detection accuracy and proactive defense.
4	Python for Cybersecurity Automation	Python Software Foundation	Demonstrated Python-based automation for security log processing and incident response.	Python	Python simplifies automation, log processing, and security scripting.
5	Cloud-Based Security Information and Event Management	Sumo Logic Inc.	Explained cloud-native SIEM architecture for centralized monitoring and analytics.	Sumo Logic Cloud SIEM	Cloud SIEM provides scalability, real-time monitoring, and centralized security visibility.
6	GeoIP-Based Threat Analysis for Security Monitoring	MaxMind / IP Geolocation Research	Used GeoIP enrichment to determine attacker locations and improve security investigations.	GeoIP Services	GeoIP provides geographical context for security events and supports incident analysis.

III. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

With the rapid increase in cyber threats, organizations generate a massive amount of security logs from servers, applications, firewalls, network devices, and cloud environments. In many organizations, these logs are monitored using traditional security tools that operate independently without centralized log management. Security analysts often rely on manual log analysis to identify suspicious activities, making the monitoring process time-consuming and inefficient.

Existing security monitoring systems generally focus on basic security mechanisms such as firewalls, antivirus software, intrusion detection systems, and periodic vulnerability assessments. Although these solutions provide an initial layer of protection, they often lack automated event correlation, real-time threat detection, threat intelligence integration, and centralized visibility across multiple security devices. As cyberattacks become more sophisticated, traditional monitoring methods struggle to detect advanced threats efficiently.

In many organizations, incident response activities such as threat investigation, report generation, alert management, and security event prioritization are performed manually. This increases the workload of Security Operations Center (SOC) analysts and may result in delayed detection and response to security incidents. The absence of automated risk assessment and standardized attack classification further limits the effectiveness of existing security monitoring systems.

DISADVANTAGES

- Manual analysis of large volumes of security logs is time-consuming.
- Lack of centralized monitoring across multiple security devices.
- Delayed threat detection and incident response.
- Limited integration with Threat Intelligence and automated attack correlation.
- Increased workload for Security Operations Center (SOC) analysts.

3.2 PROPOSED SYSTEM

The proposed system introduces a Cloud-Based Security Information and Event Management (SIEM) Analytics and Automated Incident Response solution using Python and Sumo Logic. Unlike traditional security monitoring methods, the proposed system centralizes security log collection, enriches events with threat intelligence, and automates security analysis through cloud-based monitoring. The system is designed to improve Security Operations Center (SOC) efficiency by reducing manual intervention and enabling real-time threat detection.

The proposed solution simulates various cyberattack scenarios such as brute-force attacks, SQL injection, reconnaissance scans, and suspicious login attempts using Python-generated security logs. These logs are enriched with Threat Intelligence information, GeoIP data, MITRE ATT&CK technique mapping, and dynamic risk score calculations before being transmitted to the Sumo Logic cloud platform for centralized analysis and visualization.

The system further automates several incident response activities, including Executive SOC Report generation, Incident Reports, IOC summaries, SOC email alerts, and firewall blocking recommendations. Interactive dashboards provide real-time visibility into attack trends, malicious IP addresses, risk levels, and MITRE ATT&CK techniques, enabling security analysts to detect and respond to cyber threats more efficiently.

ADVANTAGES

- Centralized cloud-based security monitoring.
- Real-time log collection and threat detection.
- Automated Threat Intelligence enrichment.
- MITRE ATT&CK framework integration.
- Automated Executive SOC and Incident Report generation.
- Improved Security Operations Center (SOC) efficiency.
- Faster incident detection and response.
- Scalable cloud-based security analytics

IV. SYSTEM OVERVIEW

The proposed system integrates cloud-based Security Information and Event Management (SIEM) with Python automation to improve security monitoring and incident response. It combines multiple technologies to collect, analyze, enrich, and visualize security events while automating several Security Operations Center (SOC) activities. The major technologies used in the proposed system are discussed below.

4.1 PYTHON

Python is a high-level, interpreted programming language widely used in automation, data processing, cybersecurity, and cloud computing. Its simplicity, extensive libraries, and cross-platform compatibility make it one of the most popular programming languages for developing security automation solutions. In this project, Python is used to simulate cyber attack scenarios, generate security logs, integrate Threat Intelligence data, calculate risk scores, generate reports, and automate incident response activities before transmitting the processed logs to the Sumo Logic cloud SIEM platform.

4.2 SUMO LOGIC

Sumo Logic is a cloud-native Security Information and Event Management (SIEM) platform that provides centralized log management, real-time monitoring, security analytics, and interactive dashboards. It enables organizations to collect and analyze security events from multiple sources through a scalable cloud infrastructure. In this project, Sumo Logic is used to receive Python-generated security logs, perform centralized analysis, visualize attack trends, monitor MITRE ATT&CK techniques, and generate dashboards for Security Operations Center (SOC) monitoring.

4.3 THREAT INTELLIGENCE AND MITRE ATT&CK FRAMEWORK

Threat Intelligence provides valuable information about malicious IP addresses, Indicators of Compromise (IOCs), and emerging cyber threats collected from trusted security sources. MITRE ATT&CK is a globally recognized knowledge base that categorizes attacker tactics and techniques based on real-world cyberattacks. In this project, Threat Intelligence is used to enrich security logs, while the MITRE ATT&CK framework maps detected attacks to standardized techniques, enabling security analysts to understand attacker behavior and improve incident investigation.

4.4 SECURITY LOG ENRICHMENT

Security log enrichment is the process of adding contextual information to raw security logs to improve threat detection and incident investigation. Instead of analyzing only the basic log information, additional details such as geographical location, threat intelligence, attack techniques, and risk scores are attached to each security event. This provides security analysts with better visibility into suspicious activities and helps prioritize critical threats.

In this project, the generated security logs are enriched using GeoIP information, Threat Intelligence, MITRE ATT&CK mapping, and dynamic risk score calculations before being transmitted to the Sumo Logic platform. GeoIP identifies the geographical location of suspicious IP addresses, while Threat Intelligence determines whether an IP address is known to be malicious. The MITRE ATT&CK framework classifies each detected activity according to standardized attacker tactics and techniques. Risk scores are then assigned based on the severity and characteristics of each event, allowing security analysts to focus on high-priority incidents.

The enriched security logs improve the efficiency of Security Operations Center (SOC) monitoring by providing meaningful context for every security event. This enables faster threat detection, better incident investigation, and more effective decision-making through centralized cloud-based security analytics.

4.5 USE CASES

The proposed Cloud-Based SIEM Analytics and Automated Incident Response system can be applied in various real-world cybersecurity environments to improve security monitoring, threat detection, and incident response. The major use cases of the proposed system are as follows:

1. Security Operations Center (SOC) Monitoring

- Enables centralized monitoring of security events from multiple systems.
- Helps SOC analysts identify and respond to threats in real time.

2. Enterprise Security Monitoring

- Collects and analyzes security logs from servers, applications, and network devices.
- Provides continuous monitoring to improve an organization's security posture.

3. Threat Hunting and Incident Investigation

- Uses Threat Intelligence and MITRE ATT&CK mapping to identify malicious activities.
- Assists security analysts in investigating Indicators of Compromise (IOCs) and attack patterns.

4. Automated Incident Response

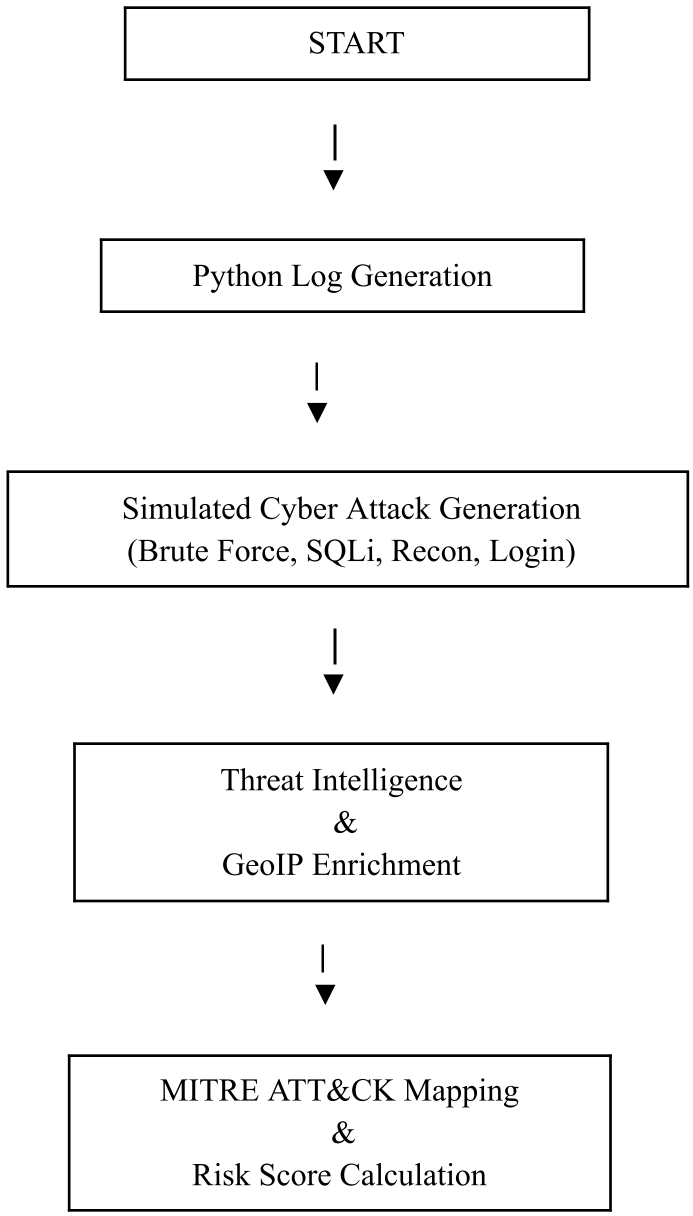
- Automatically generates Executive SOC Reports, Incident Reports, IOC summaries, and SOC email alerts.
- Supports faster response to critical security incidents while reducing manual effort.

5. Cybersecurity Education and Research

- Provides a practical learning platform for students and researchers to understand cloud-based SIEM, SOC workflows, log analysis, and incident response automation.

V. METHODOLOGY

5.1 WORKFLOW



↓

Upload Logs to
Sumo Logic Cloud SIEM

↓

Dashboard Visualization
&
Security Event Analysis

↓

Automated Incident Response

↓

Reports • Alerts • Firewall Block

↓

END

5.1 SYSTEM WORKFLOW

The proposed methodology follows an automated workflow for detecting, analyzing, and responding to cyber threats using a cloud-based Security Information and Event Management (SIEM) platform. Initially, Python generates simulated security logs representing various cyber attack scenarios such as brute-force attacks, SQL injection, reconnaissance scans, and suspicious login attempts. The generated logs are then enriched with Threat Intelligence information, GeoIP data, MITRE ATT&CK mappings, and dynamic risk score calculations to provide additional context for security analysis.

The enriched security logs are transmitted to the Sumo Logic cloud SIEM platform, where they are collected, analyzed, and visualized through interactive dashboards. Based on the detected security events, the system automatically generates Executive SOC Reports, Incident Reports, IOC summaries, SOC email alerts, and firewall blocking recommendations. This automated workflow improves threat visibility, reduces manual effort, and enables efficient Security Operations Center (SOC) operations.

5.2 SECURITY LOG GENERATION

Security log generation is the first stage of the proposed system. Since obtaining real organizational security logs is difficult due to privacy and security restrictions, Python is used to generate simulated security logs that closely resemble real-world security events. These logs provide a safe environment for testing, monitoring, and evaluating the cloud-based SIEM platform.

The generated logs simulate various cyber attack scenarios, including brute-force attacks, SQL injection attempts, reconnaissance scans, suspicious login activities, and normal user behavior. Each log entry contains important information such as timestamp, source IP address, destination endpoint, attack type, event status, severity level, and other security-related attributes. Randomized values are generated to create realistic and diverse security events.

The generated logs are then prepared for further processing, where they are enriched with Threat Intelligence, GeoIP information, MITRE ATT&CK mappings, and risk scores before being transmitted to the Sumo Logic cloud SIEM platform for centralized analysis.

Purpose of Security Log Generation

- Simulates real-world cyber attack scenarios.
- Generates structured security logs using Python.
- Provides data for testing cloud-based SIEM analytics.
- Enables realistic SOC monitoring without exposing real organizational data.
- Forms the input for Threat Intelligence enrichment and automated incident response.

5.3 THREAT INTELLIGENCE INTEGRATION

Threat Intelligence Integration enhances raw security logs by providing additional contextual information about suspicious activities and malicious IP addresses. Instead of analyzing basic log data alone, the proposed system enriches each security event with external threat information to improve the accuracy of threat detection and incident investigation. This process enables security analysts to identify known malicious sources and prioritize high-risk security events more effectively.

In this project, the generated security logs are compared with Threat Intelligence sources to determine whether an IP address has been previously identified as malicious. The logs are further enriched using GeoIP information to identify the geographical location of the source IP address. Each event is then mapped to the appropriate MITRE ATT&CK technique, providing a standardized understanding of attacker behavior. Based on the severity of the attack, threat intelligence results, and MITRE mapping, a dynamic risk score is assigned to every security event.

The enriched logs are then forwarded to the Sumo Logic cloud SIEM platform for centralized analysis and visualization. This enrichment process significantly improves threat visibility, supports faster incident investigation, and enables Security Operations Center (SOC) analysts to respond more efficiently to potential cyber threats.

Functions of Threat Intelligence Integration

- Identifies known malicious IP addresses.
- Enriches logs with GeoIP location information.
- Maps attacks to the MITRE ATT&CK framework.
- Assigns dynamic risk scores based on event severity.
- Improves threat detection and incident investigation.
- Provides enriched security events for cloud SIEM analysis.

5.4 CLOUD SIEM ANALYTICS

Cloud SIEM Analytics is the core component of the proposed system that enables centralized security monitoring and real-time analysis of security events. After the security logs are enriched with Threat Intelligence, GeoIP information, MITRE ATT&CK mappings, and risk scores, they are transmitted to the Sumo Logic cloud SIEM platform for further processing. The platform collects, stores, and analyzes the incoming security events to identify suspicious activities and potential cyber threats.

In this project, Sumo Logic performs real-time log analysis by correlating security events from the generated logs. Interactive dashboards provide visual representations of attack trends, malicious IP addresses, security event severity, and MITRE ATT&CK techniques. These dashboards help Security Operations Center (SOC) analysts quickly identify abnormal activities, investigate incidents, and monitor the overall security posture of the system.

Cloud SIEM analytics also supports advanced search queries, event filtering, and centralized log management, allowing security analysts to investigate incidents efficiently. By providing real-time visibility into security events, the proposed system improves threat detection, reduces investigation time, and enhances overall incident response capabilities.

Functions of Cloud SIEM Analytics

- Collects and stores enriched security logs.
- Performs centralized log analysis.
- Visualizes security events through interactive dashboards.
- Supports real-time monitoring and event correlation.
- Improves incident investigation and threat detection.
- Provides centralized Security Operations Center (SOC) monitoring.

5.5 AUTOMATED INCIDENT RESPONSE

Automated Incident Response is the final stage of the proposed system, where detected security events are automatically processed to reduce manual intervention and improve response time. Based on the severity of each security event, the system generates multiple security reports and notifications that assist Security Operations Center (SOC) analysts in responding to cyber threats effectively.

In this project, Python automation generates Executive SOC Reports, Incident Reports, Indicators of Compromise (IOC) summaries, and SOC email alerts based on the analyzed security events. The system also recommends firewall blocking actions for malicious IP addresses identified through Threat Intelligence. These automated processes enable faster incident handling while reducing the workload of security analysts.

By automating routine SOC activities, the proposed system improves operational efficiency, accelerates incident response, and provides a structured approach for managing cybersecurity incidents in cloud-based environments.

Functions of Automated Incident Response

- Generates Executive SOC Reports automatically.
- Creates Incident Reports and IOC summaries.
- Sends automated SOC email alerts.
- Recommends firewall blocking for malicious IP addresses.
- Reduces manual effort during incident handling.
- Improves response time and SOC efficiency.

VI. RESULT AND DISCUSSION

6.1 SECURITY LOG GENERATION

Step 1: Execute the Python Program

The Python program is executed to generate simulated security logs representing various cyber attack scenarios such as brute-force attacks, SQL injection attempts, reconnaissance scans, suspicious login activities, and normal user behavior. These logs act as the input for the Cloud SIEM platform.

```
        country_count[country] = country_count.get(country, 0) + 1

    print("\n========================================")
    print("📊 SOC TREND ANALYSIS")
    print("========================================")

    print("\nAttack Types")
    for attack, count in attack_count.items():
        print(f"{attack:<25} {count}")

    print("\nSeverity Distribution")
    for severity, count in severity_count.items():
        print(f"{severity:<25} {count}")

    print("\nTop Source Countries")
    for country, count in sorted(country_count.items(),
                                 key=lambda x: x[1],
                                 reverse=True):
        print(f"{country:<25} {count}")

    print("========================================\n")

# -------------------------
```

Step 2: Generated Security Logs

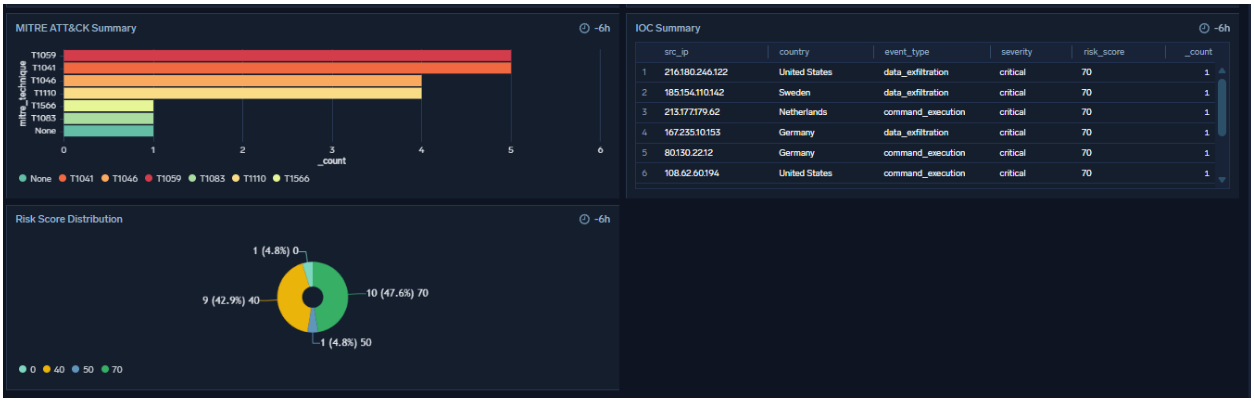
The generated logs contain essential security information such as timestamp, source IP address, attack type, severity level, destination endpoint, and event status. These logs are then prepared for Threat Intelligence enrichment before being transmitted to the Sumo Logic Cloud SIEM platform.

```
Total Security Events : {total_logs}

Critical Threats      : {critical}

Blocked Attacks       : {blocked}

Countries Involved    : {countries}

Threat Intelligence   : Enabled

GeoIP Enrichment      : Enabled

MITRE ATT&CK Mapping  : Enabled

Risk Scoring          : Enabled

Email Alerting        : Enabled

Incident Reporting    : Enabled

Automatic Blocking    : Enabled

Overall SOC Status    : OPERATIONAL
```

6.2 THREAT INTELLIGENCE DASHBOARD

Step 1: Threat Intelligence Integration

The generated security logs are enriched using Threat Intelligence data to identify malicious IP addresses and suspicious activities. This process provides additional context to security events and improves the accuracy of threat detection.

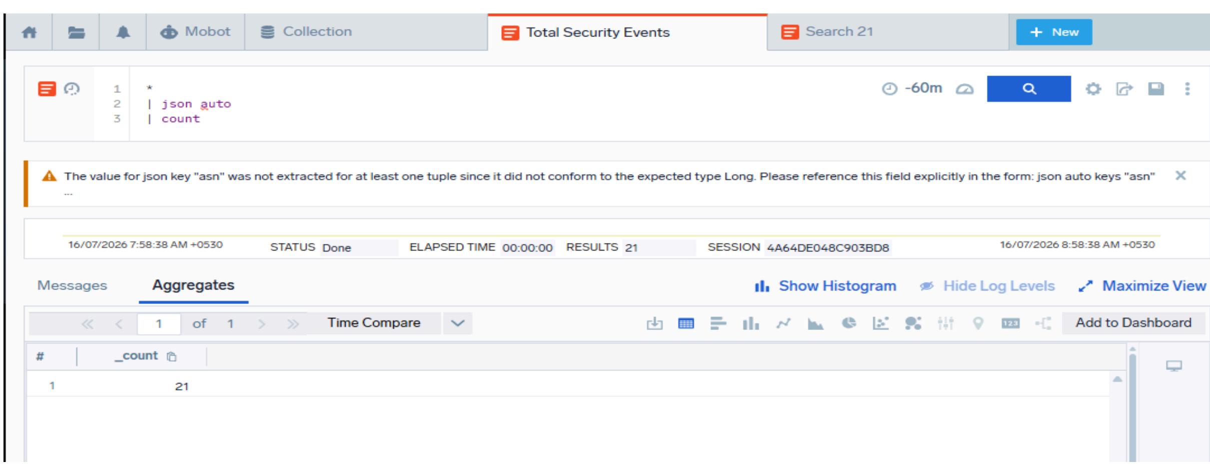


The Threat Intelligence dashboard displays malicious IP addresses, attack details, and security events received from the Python application. This enables Security Operations Center (SOC) analysts to quickly identify and investigate potential cyber threats.

6.3 TOTAL SECURITY EVENTS

Step 1: Security Event Monitoring

The Cloud SIEM platform continuously receives and processes security logs generated by the Python application. Every simulated attack and normal activity is recorded as a security event.

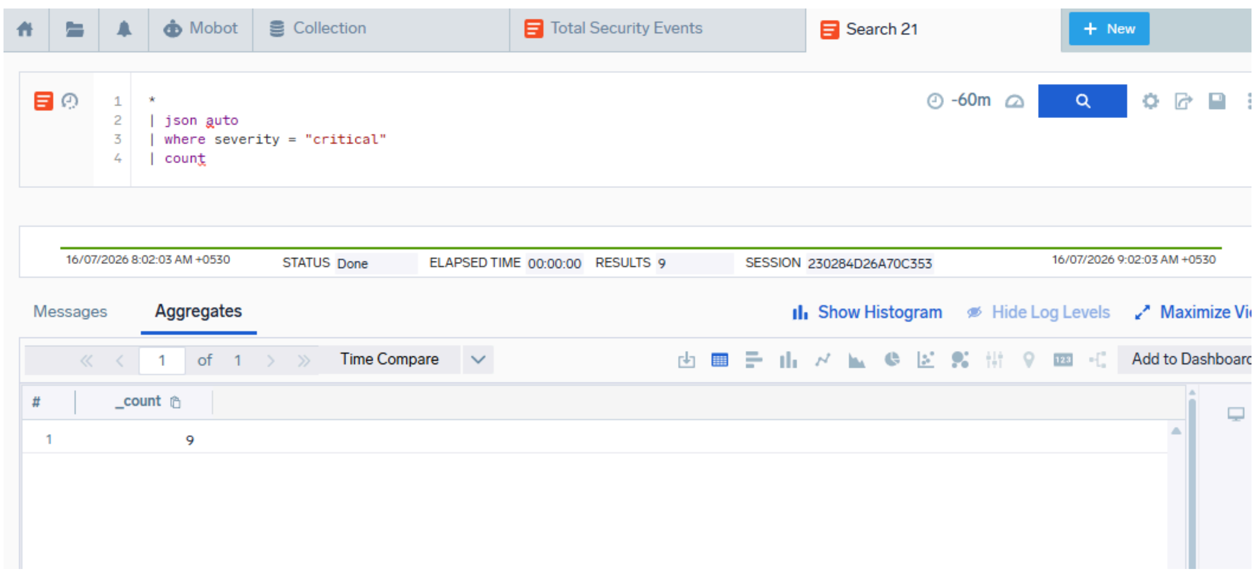


The Total Security Events dashboard provides the overall count of processed security events. It offers analysts a quick overview of system activity and helps monitor the volume of incoming security logs.

6.4 CRITICAL THREAT DETECTION

Step 1: Critical Threat Identification

The proposed system analyzes each security event based on predefined detection rules and risk scores. Events with high severity are classified as critical threats requiring immediate attention.

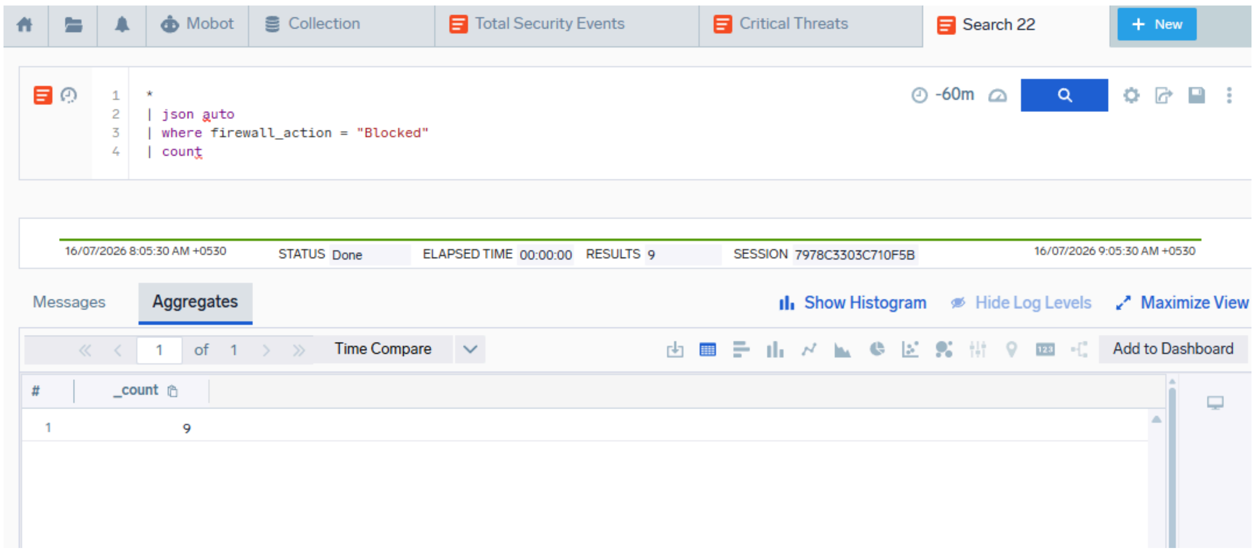


The dashboard highlights all critical threats separately, allowing Security Operations Center (SOC) analysts to prioritize incident investigation and initiate rapid response procedures.

6.5 BLOCKED ATTACKS

Step 1: Automatic Attack Blocking

The proposed system identifies malicious activities based on predefined detection rules and threat intelligence information. High-risk attacks are automatically marked for blocking to prevent further communication from malicious IP addresses.

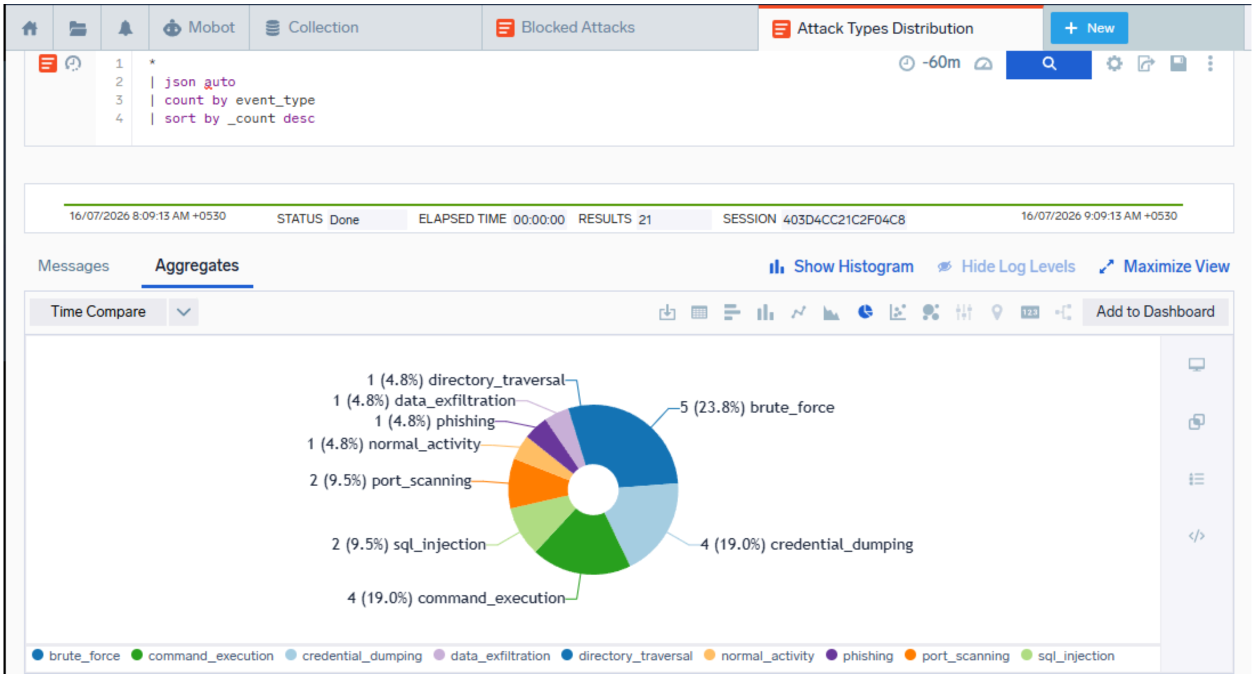


The dashboard displays the total number of blocked attacks detected during security monitoring. This helps SOC analysts verify that malicious activities have been successfully identified and appropriate defensive actions have been initiated.

6.6 ATTACK TYPE DISTRIBUTION

Step 1: Attack Classification

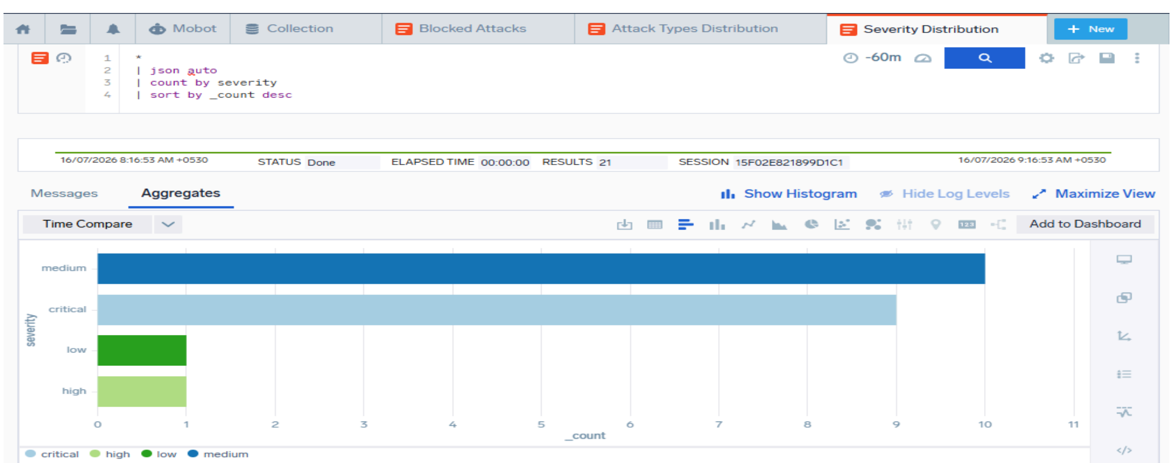
The detected security events are classified into different attack categories such as brute-force attacks, SQL injection, reconnaissance scans, suspicious login attempts, and normal user activities



The Attack Type Distribution chart provides a graphical representation of the frequency of different attack types. This visualization helps security analysts identify the most common attack patterns and understand the overall threat landscape.

6.7 SEVERITY DISTRIBUTION

Step 1: Severity Classification

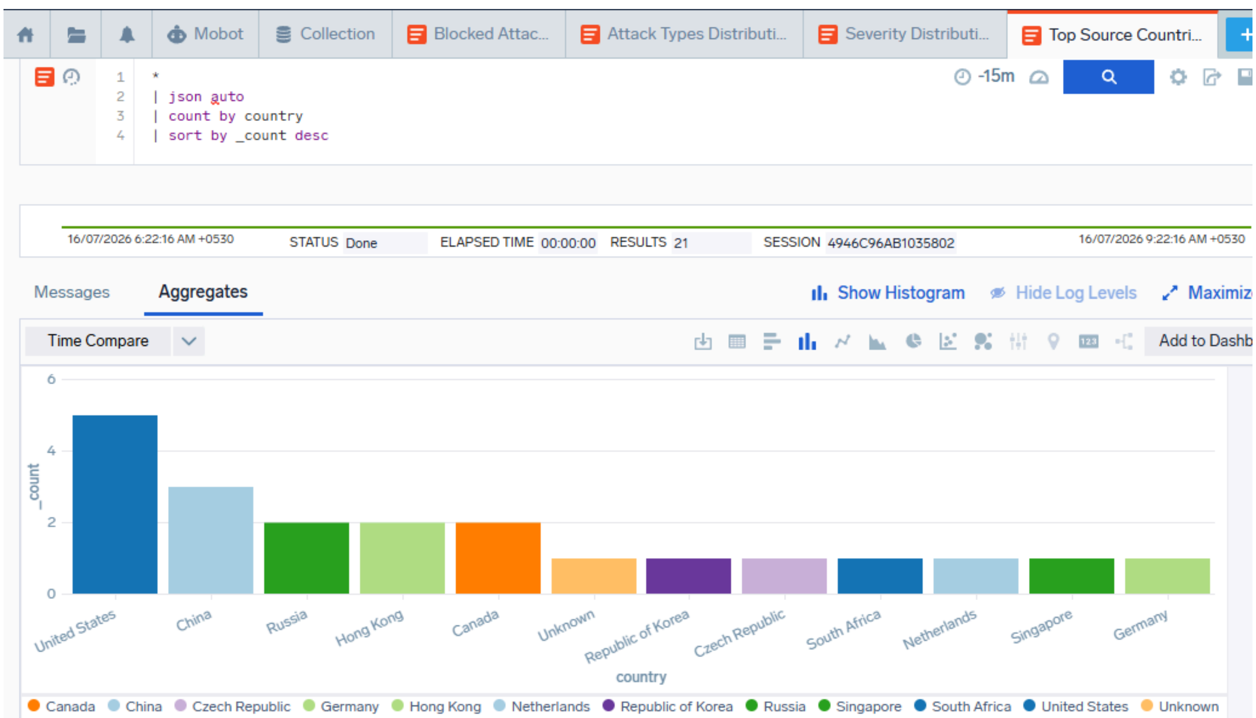


Each detected security event is assigned a severity level based on its potential impact and calculated risk score.The Severity Distribution chart categorizes security events into Low, Medium, High, and Critical levels. This enables analysts to prioritize high-risk incidents and allocate resources effectively for incident response.

6.8 TOP SOURCE COUNTRIES

Step 1: GeoIP Analysis

The proposed system uses GeoIP information to determine the geographical location of source IP addresses. This helps identify the origin of security events and detect suspicious activities from different regions.

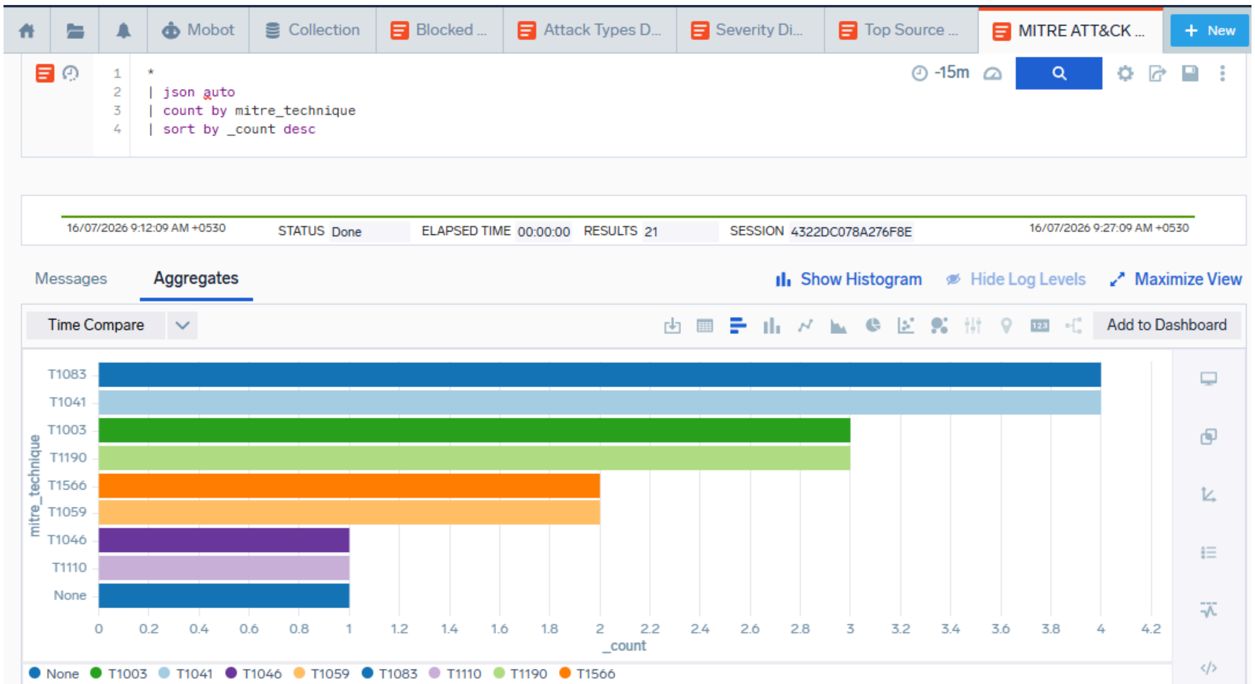


The Top Source Countries dashboard displays the countries from which security events originated. This visualization assists SOC analysts in identifying geographical attack patterns and monitoring regional threat activity.

6.9 MITRE ATT&CK TECHNIQUES

Step 1: MITRE ATT&CK Mapping

Detected security events are mapped to the MITRE ATT&CK framework based on the attack techniques identified during log analysis.

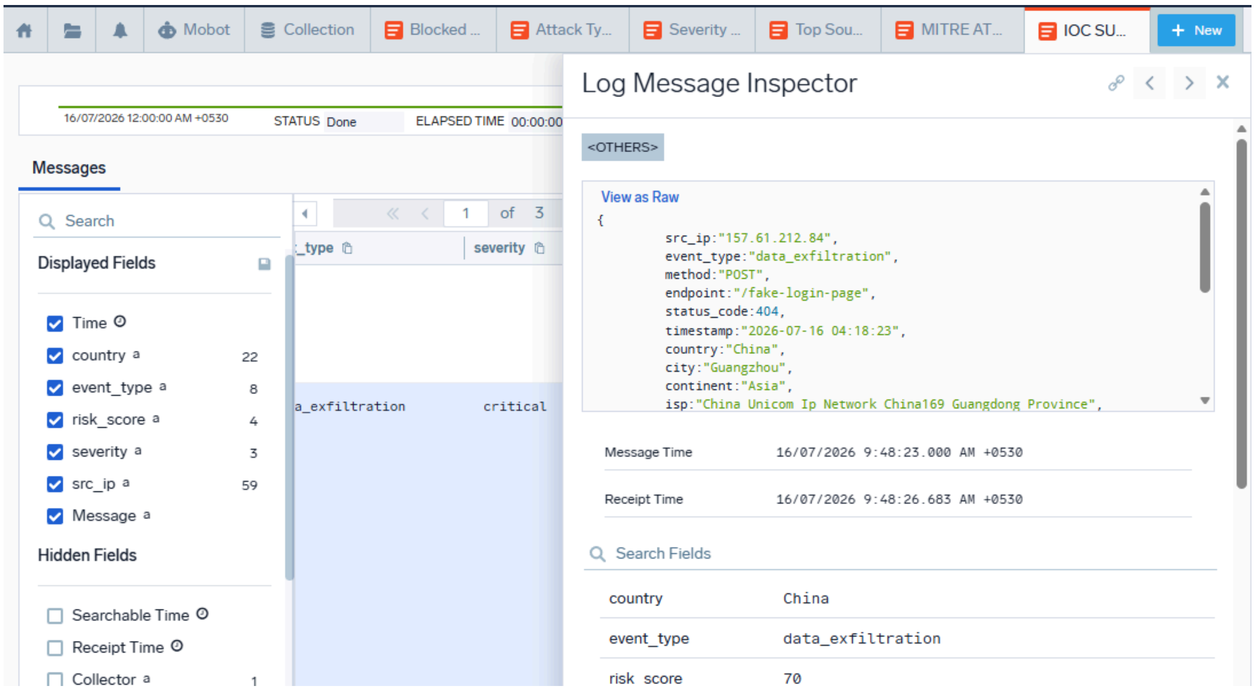


The MITRE ATT&CK dashboard provides a standardized view of attacker tactics and techniques. This enables analysts to understand attacker behavior and improve incident investigation and response.

6.10 IOC SUMMARY

Step 1: Indicator of Compromise (IOC) Identification

The system extracts Indicators of Compromise (IOCs) such as malicious IP addresses, attack signatures, and suspicious activities from the processed security logs.

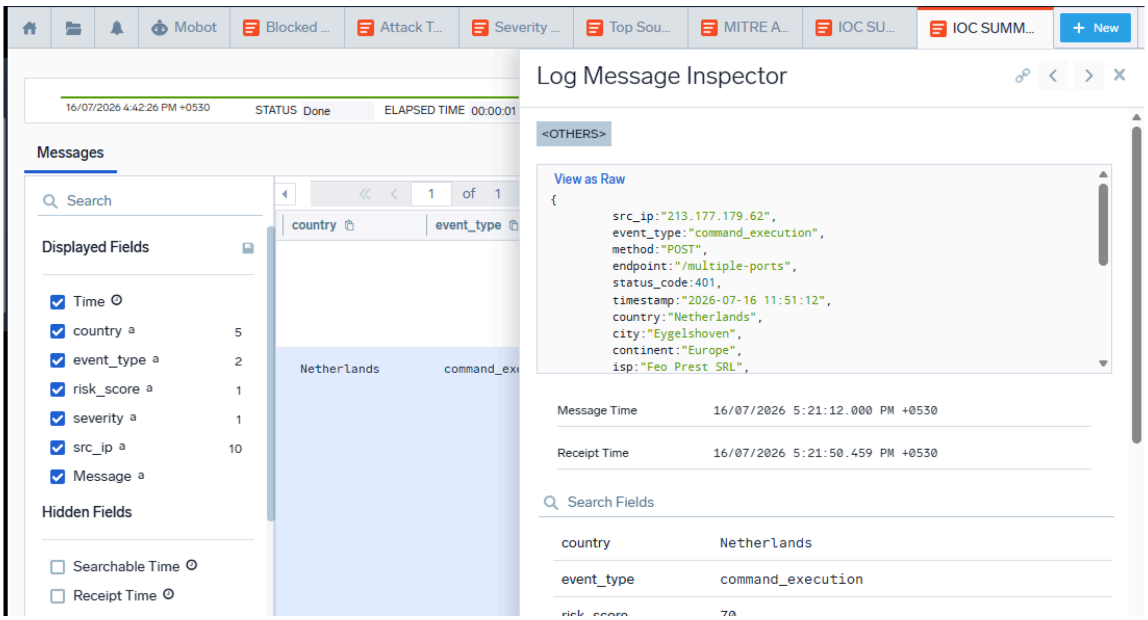


The IOC Summary provides a consolidated view of detected indicators of compromise, allowing security analysts to investigate malicious activities and support threat-hunting operations.

6.11 IOC SUMMARY FOR BLOCKED THREATS

Step 1: Blocked IOC Identification

The system maintains a separate summary of blocked malicious IP addresses and security events that triggered automatic defensive actions.



This summary helps analysts verify blocked threats, review mitigation actions, and maintain records of prevented malicious activities for future analysis.

6.12 EXECUTIVE SOC REPORT

Step 1: Executive Report Generation

The proposed system automatically generates an Executive SOC Report after analyzing the processed security events. The report provides a high-level summary of the organization's security posture, making it easier for management to understand overall security performance.

```
Total Security Events : {total_logs}
Critical Threats      : {critical}
Blocked Attacks       : {blocked}
Countries Involved    : {countries}
Threat Intelligence   : Enabled
GeoIP Enrichment      : Enabled
MITRE ATT&CK Mapping  : Enabled
Risk Scoring          : Enabled
Email Alerting        : Enabled
Incident Reporting    : Enabled
Automatic Blocking    : Enabled
Overall SOC Status    : OPERATIONAL
```

The Executive SOC Report summarizes total security events, critical threats, blocked attacks, attack distribution, and overall incident statistics. It assists management in reviewing security performance and making informed security decisions.

6.13 MITRE SUMMARY

Step 1: MITRE Summary Generation

The system generates a summary of all detected MITRE ATT&CK techniques identified during security event analysis.

```
========================================
      MITRE ATT&CK SUMMARY
========================================
T1041          5
T1059          5
T1046          4
T1110          4
T1083          1
T1566          1
None           1
========================================
```

The report provides a consolidated view of attacker tactics and techniques, helping security analysts understand attack patterns and strengthen incident investigation.

6.14 LIVE SOC DASHBOARD

Step 1: Real-Time Dashboard Monitoring

The Live SOC Dashboard continuously displays incoming security events collected from the Python-generated logs.

```
========================================
        LIVE SOC DASHBOARD
========================================
Total Events        : 21
Critical Threats    : 10
Blocked Attacks     : 10
Countries Involved  : 8
SOC Status          : ACTIVE
========================================
```

The dashboard provides real-time visibility into attack trends, security alerts, and system activity, enabling Security Operations Center (SOC) analysts to monitor threats efficiently.

6.15 SOC EMAIL ALERT

Step 1: Email Alert Generation

The proposed system automatically generates email alerts whenever critical or high-risk security events are detected.

```
========================================
🚨 SOC EMAIL ALERT
========================================
Subject : CRITICAL SECURITY ALERT
Source IP      : 213.177.179.62
Country        : Netherlands
City           : Eygelshoven
Attack Type    : command_execution
MITRE ATT&CK   : T1059
Severity       : critical
Risk Score     : 70
Firewall Action: Blocked
========================================
```

The email alert contains important information such as attack type, source IP address, severity level, and recommended actions. This enables security analysts to respond quickly to potential cyber threats through the notification.

6.16 INCIDENT REPORT

Step 1: Incident Report Generation

```
========================================
📄 INCIDENT REPORT
========================================
Source IP        : 213.177.179.62
Country          : Netherlands
City             : Eygelshoven
Attack Type      : command_execution
MITRE Technique  : T1059
Severity         : critical
Risk Score       : 70
Firewall Action  : Blocked
Status           : Incident Logged
========================================
```

An Incident Report is automatically created for every detected security incident.The Incident Report includes attack details, severity, MITRE ATT&CK mapping, risk score, source IP address, and recommended mitigation actions. It supports detailed incident investigation and documentation.

6.17 AUTOMATIC FIREWALL BLOCKING

Step 1: Firewall Blocking Recommendation

The system identifies malicious IP addresses using Threat Intelligence and recommends firewall blocking actions.

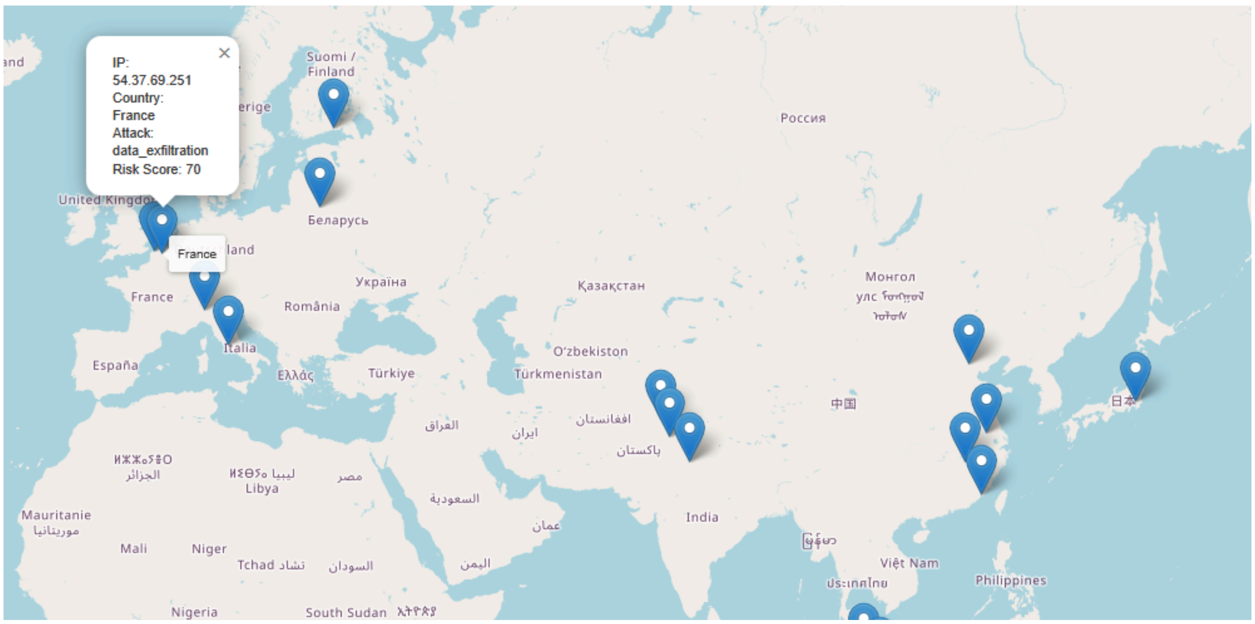
```
==============================
[SOC RESPONSE]
High-Risk Threat Detected
IP: 213.177.179.62
Risk Score: 70
Firewall Action: BLOCKED
==============================
```

The Automatic Blocking report displays blocked malicious IP addresses and the corresponding security events. This helps prevent repeated attacks and strengthens the organization's security posture.

6.18 SOC ATTACK MAP

Step 1: Attack Map Visualization

The system visualizes the geographical origin of cyber attacks using GeoIP information.The SOC Attack Map displays the countries from which attacks originated, enabling security analysts to identify regional attack patterns and gain better situational awareness during security monitoring.



VII. CONCLUSION

CONCLUSION

The project "Cloud-Based SIEM Analytics and Automated Incident Response Using Python" successfully demonstrates the implementation of a cloud-based Security Information and Event Management (SIEM) solution for monitoring and analyzing cybersecurity events. By integrating Python automation with the Sumo Logic Cloud SIEM platform, the system effectively simulates various cyber attack scenarios, processes security logs, enriches them with Threat Intelligence and GeoIP information, maps attacks to the MITRE ATT&CK framework, and calculates dynamic risk scores.

The proposed system provides centralized security monitoring through interactive dashboards, enabling Security Operations Center (SOC) analysts to identify suspicious activities and prioritize critical threats. Automated generation of Executive SOC Reports, Incident Reports, IOC summaries, SOC email alerts, and firewall blocking recommendations significantly reduces manual effort and improves incident response efficiency.

Overall, the project demonstrates how cloud-based SIEM platforms combined with automation can enhance threat detection, accelerate incident response, and strengthen organizational cybersecurity by providing real-time visibility into security events.

VIII. FUTURE ENHANCEMENT

Future improvements can further enhance the capabilities of the proposed Cloud-Based SIEM system. Some possible enhancements include:

- **Real-Time Log Collection**
  Integrate the system with live enterprise devices such as firewalls, routers, servers, and operating systems instead of using simulated security logs.
- **Machine Learning-Based Threat Detection**
  Implement machine learning algorithms to identify unknown attack patterns, anomalies, and zero-day cyber threats.
- **SOAR Integration**
  Integrate the system with Security Orchestration, Automation, and Response (SOAR) platforms to automate incident investigation and response workflows.
- **Real-Time Threat Intelligence Feeds**
  Integrate live Threat Intelligence sources such as AlienVault OTX, AbuseIPDB, VirusTotal, and MISP to improve threat detection accuracy.
- **Automated Ticketing System**
  Connect the system with incident management tools such as ServiceNow or Jira to automatically generate and assign security incident tickets.
- **Multi-Cloud SIEM Support**
  Extend the solution to support multiple cloud SIEM platforms such as Splunk, Microsoft Sentinel, IBM QRadar, and Google Chronicle.

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